IS507 project proposal (Team 17)

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Motivation and problem

Regression predicts an observed outcome from measured features. A treatment-effect question is different: for people with a given feature profile, how would the outcome change under treatment compared with no treatment? For one person, only the outcome under the treatment actually received is observed. The other outcome is a counterfactual, so individual treatment effects cannot be directly labeled in ordinary data.

I chose this problem because it connects my interest in causal inference with regression and model evaluation. Estimating effects across feature-defined groups could help researchers understand who may benefit more from an intervention and guide further study. A simulation comparison can assess estimators under controlled conditions; by itself it cannot justify a real clinical or policy decision.

The target is the conditional average treatment effect (CATE), a function of pre-treatment features:

`tau(x) = E[Y(1) - Y(0) | X = x].`

Here, `X` is a person's measured feature vector; `A` is a binary indicator for treatment received; and `Y` is the observed outcome. `Y(1)` and `Y(0)` denote potential outcomes under treatment and no treatment. The learner receives rows `(X, A, Y)` and returns an estimated function `tau_hat(x)`. It does not observe both potential outcomes for a person. Under consistency, conditional unconfoundedness, and overlap, the CATE is identifiable from these observed data. The study will make these assumptions hold by design in simulation.

Related work and candidate learners

The assigned paper, Curth and van der Schaar (2021), organizes nonparametric CATE estimators, studies their theoretical properties, and compares neural-network implementations. It discusses the S-, T-, and X-learners associated with Künzel et al. (2019), and doubly robust (DR) learning associated with Kennedy (2020). These summaries follow the assigned paper; I have not independently reviewed those cited papers.

The baseline will be a T-learner implemented with two separate outcome neural networks (TNet): fit one outcome model among treated rows and another among untreated rows, then subtract their predictions. Each model minimizes squared outcome prediction error. This is a useful reference because it directly estimates the two conditional outcome means without weighting or constructed effect labels.

I will compare this baseline with three two-step learners from the assigned paper:

- **Regression adjustment (RA):** use outcome models to fill in the unobserved treatment-group mean and construct a row-level effect target.
- **Propensity weighting (PW):** weight outcomes by the estimated probability of the treatment received. Small probabilities can produce noisy, large weights.
- **Doubly robust (DR):** combine outcome-model predictions with propensity-weighted residuals. Its theoretical robustness does not guarantee the smallest finite-sample error in every setting.

The comparison asks when these adjustments improve CATE estimation over the simpler T-learner, rather than proposing a new estimator.

Proposed data

I will generate synthetic data with one person per row. Each row will contain 25 continuous, Gaussian-distributed pre-treatment features, a binary treatment `A`, and a continuous outcome `Y`. The proposed development sample sizes are 500 and 2,000 rows, with a separate test set of 500 rows. Treatment

probabilities will be kept between 0.1 and 0.9 to avoid near-deterministic assignment. I will vary the CATE function between simple and more complex forms. Because the simulator specifies both potential outcomes, it also knows the true `tau(x)` for evaluation; the learners themselves will only receive `(X, A, Y)`.

This is a proposed simulation, not a collected dataset or an exact reproduction of the paper. The design has no text, images, or missing feature entries. After generation, I will check feature missingness, treatment counts, and treatment-probability ranges rather than assume the generated data passed those checks. The main data difficulty is that each row lacks an observed outcome under the other treatment. With limited samples, I also expect overfitting, noisy RA/PW/DR targets, and high variability from inverse weights.

Proposed data-generating process

To make the proposal reproducible, I will start with independent standard-normal features, `X_j ~ Normal(0, 1)`. The first three features carry the designed signal; the remaining 22 are noise features. This lets the study test whether learners can find effect modifiers when many recorded columns are irrelevant. Treatment assignment will depend on observed features through a logistic link:

```
e(x) = clip(sigmoid(0.5 x_1 - 0.5 x_2), 0.1, 0.9)
A | X=x ~ Bernoulli(e(x)).
```

The clipping step keeps both treatment choices possible for each profile. Since assignment depends on measured `X`, this creates observed confounding while retaining overlap. For outcomes, I will use a baseline mean `mu_0(x) = x_1 + 0.5 x_2` and define `Y(a) = mu_0(X) + a tau(X) + epsilon`, where `epsilon` is mean-zero Gaussian noise. I will compare a simple effect `tau_simple(x) = 1 + 0.5 x_1` with a nonlinear effect `tau_complex(x) = 1 + 0.5 sin(x_1) + 0.25 x_2 x_3`. The observed outcome is `Y = A Y(1) + (1-A)Y(0)`, and the simulator knows `tau(x)` exactly. The numerical coefficients and noise scale will be fixed in the experiment configuration before the runs and reported with the code.

This first design isolates sample size and effect-function complexity while keeping the feature distribution, assignment mechanism, baseline outcome, and noise rule comparable. It does not cover every form of misspecification, feature correlation, or hidden confounding. Those are limitations of the initial experiment, rather than claims the proposed results can resolve. The logistic link above generates assignments; during model fitting, the candidate learners will still estimate treatment probabilities from observed training rows.

Training and evaluation plan

For each sample-size and effect-complexity setting, I will use 10 random seeds. Within each seed, all learners will use the same generated observations and data splits so the comparison is paired. I will split development rows into 80% training and 20% validation, keeping the independent test rows untouched. Feature scaling will be fit using training rows only.

The outcome networks will minimize squared error; the treatment-probability network will minimize binary cross-entropy. For two-step learners, I will use two-fold cross-fitting within the training data: fit first-stage models on one fold, create targets for the other fold, then swap folds. I will regress the resulting out-of-fold targets on `X` using squared error. Learners will use comparable neural-network capacity and the same validation-tuning budget. Since this is a simulation, validation CATE RMSE can be computed using the known true CATE; the test set will be used only for final evaluation.

The same split indices will be shared across learners within a seed. Validation may select among configurations, but no choice will be made from test performance. The planned grid has two development-sample sizes by two effect forms, with ten seeds per cell. Thus there are 40 paired simulation runs, each with a 500-row test set. Within each run, every method sees the same training and validation

rows. This pairing reduces noise in method-to-method comparisons because a learner is not rewarded simply for receiving an easier data draw.

The primary metric is CATE root mean squared error (RMSE) on the 500 test rows:

`RMSE_CATE = sqrt(mean_j[(tau_hat(X_j) - tau(X_j))^2]).`

Lower values indicate predictions closer to the simulator's true CATE. I will report the mean and standard error across the 10 runs, along with training time. Outcome prediction error is a different metric and will not be treated as a substitute for CATE error.

The main analysis will report each learner's RMSE within each of the four design cells, then compare it with the T-learner baseline. I will examine whether the ranking changes with sample size or effect complexity, and whether PW shows more run-to-run variability when estimated treatment probabilities are small. These are questions and expected failure modes, not predetermined conclusions. If one method has lower average error but much higher variability, both features will be discussed rather than collapsing them into a single winner label.

Expected contribution and limitations

The planned result is a reproducible comparison showing how sample size and effect complexity relate to the performance of RA, PW, and DR relative to the T-learner. The assigned paper's adapted IHDP experiment illustrates why a universal winner should not be assumed: for its TNet rows, RA and plug-in outperform DR on reported hold-out CATE RMSE, while these are the authors' results and not results of this proposed study.

Simulation provides a known target for evaluation, but real observational data do not reveal the true CATE directly. Unmeasured confounding, violations of overlap, and distribution shift would limit conclusions about real interventions. The intended impact is methodological guidance for further study, not a treatment recommendation.

There are additional limits to interpretation. The simulator uses a known outcome mechanism and a known CATE, while real studies must estimate nuisance functions from noisy observations. The first design also makes treatment assignment depend only on observed features, so it cannot assess bias from an omitted confounder. Neural-network comparisons can depend on tuning and optimization, which is why capacity and tuning budget will be held comparable and runtime will be reported. A result favoring one learner in this grid would support a conditional recommendation for these settings; it would not establish that learner as best for all populations or applications.

References

Curth, A., and van der Schaar, M. (2021). "Nonparametric Estimation of Heterogeneous Treatment Effects: From Theory to Learning Algorithms." *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, PMLR 130, 1810–1819.

Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. (2019). "Metalearners for estimating heterogeneous treatment effects using machine learning." *Proceedings of the National Academy of Sciences*, 116(10), 4156–4165. (As summarized in the assigned paper.)

Kennedy, E. H. (2020). "Towards Optimal Doubly Robust Estimation of Heterogeneous Causal Effects." *Electronic Journal of Statistics*, 14(2), 3008–3044.